FLAGSHIP 06 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-TEC/W

Semiconductor Water Cooling Vest · 80W Active Liquid + Dual Airflow Chiller

80W Active Thermoelectric Chiller. Closed-Loop Liquid + Dual Airflow.

An active wearable cooling device powered by solid-state thermoelectric cooling (TEC) chips and a closed-loop water circulation system. Delivers steady, multi-level adjustable torso cooling within 2 minutes of startup. Integrates a quiet rear turbine exhaust fan and dual chest air-supply fans for all-day facial and torso dryness in firefighting, construction, power maintenance, and sanitation.

LC Frost-TEC/W · OFFICIAL HARDWARE VIEW



01 Precision Active Temperature Regulation

Thermoelectric chiller actively draws heat away from circulating liquid, overcoming passive melting limitations.

02 Dual-Zone Forced-Air Evaporation

Rear turbine expels heat while dual front fans direct air toward the neck and face to accelerate sweat evaporation.

03 Extended Field Battery Life

24V lithium power supply delivers 1-2 hours on a single pack and up to 4 hours with dual batteries.

04 Leak-Proof Sealed Liquid Channels

Industrial TPU tubing resists high pressures, twisting, and rough site conditions without leaking.

KEY METRIC 01
80W
Active Thermoelectric Cooling Capacity

KEY METRIC 02
2 min
Fast Thermal Stabilization Time

KEY METRIC 03
24V
High-Efficiency Chiller Architecture

KEY METRIC 04
2-4 Hours
Continuous Shift Battery Endurance

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



80W TEC CHILLER · 2-MIN STABILIZATION

Dorsal 80W Solid-State TEC Chiller & Turbine Exhaust

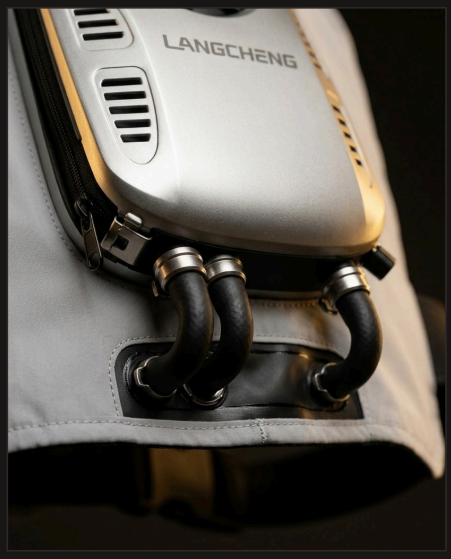
Mounted ergonomically on the back, the silver-anodized 80W thermoelectric module chills internal circulating water within 2 minutes while a quiet radial turbine exhausts rejected heat away from the operator.



DUAL ANTERIOR UPDRAFT FANS

Dual Chest Forced-Air Supply Fans

Integrated front pocket blowers channel fresh airflow upward across the chest, neck, and face, combining internal conductive liquid cooling with external convective drying.



CLOSED-LOOP TPU FLUIDIC LINER

Medical/Industrial TPU Closed-Loop Water Liner

Serpentine TPU micro-channels distribute chilled water evenly across the core and may be paired with supplemental PCM cooling packs for extreme thermal spikes.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **Antistatic FR V-Neck Shell [Anterior Flame-Retardant Chassis]**
   Soft, protective industrial grey fabric resistant to heat and static hazards.

2. **Side Compression Straps [Bilateral Thermal Contact Tensioners]**
   Ergonomic tensioners pull cooling water channels tight against the torso.

3. **Rear 80W TEC Chiller Module [Dorsal 24V / 80W Thermoelectric Core]**
   Sleek silver casing with black turbine fan stabilizes water temperatures within 2 minutes.

4. **Dual Front Air-Supply Pockets [Anterior Updraft Air-Supply Modules]**
   Front utility pockets hold batteries while driving cooling airflow up toward the chin and chest.

FIELD DEPLOYMENT | 500KV SUBSTATION TRANSFORMER OVERHAUL | SUMMER PEAK LOAD



High-Voltage Power Grid Maintenance, Firefighting & Heavy Construction

When power grid crews and firefighters operate in 40°C+ radiant heat, passive ice packs can melt before a complex maintenance task finishes. The LC Frost-TEC/W combines 80W of active thermoelectric liquid chilling with dual chest fans that sweep cooling airflow across the neck and face - sustaining peak cognitive alertness for 2 to 4 hours.

80W (24V / 100W Rated)	2 Minutes	2-4 Hours (Dual Battery)
Active Cooling Output	Loop Stabilization	Continuous Endurance

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-TEC/W (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-TEC/W
IDENTITY & COMPLIANCE	**Color**	Technical Grey (customizable)
IDENTITY & COMPLIANCE	**Target Users**	Firefighting, public security, power grid, construction, sanitation in extreme heat
PERFORMANCE & MECHANICS	**Cooling Capacity**	80W
PERFORMANCE & MECHANICS	**Rated Voltage**	24V
PERFORMANCE & MECHANICS	**Rated Power**	100W
PERFORMANCE & MECHANICS	**Cooling Response Time**	Stable in 2 min
PERFORMANCE & MECHANICS	**Battery Life**	1-2 hours single battery (2-4 hours continuous system endurance)
MATERIALS & ERGONOMICS	**Outer Fabric**	Antistatic and flame-retardant fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	Medical/industrial TPU water-channel liner
DIMENSIONS & WARRANTY	**Size**	One Size Fits All
DIMENSIONS & WARRANTY	**Net Weight**	< 3kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Supplemental PCM cooling packs
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-TEC/W

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware